For operating time and price per kWh , the least electricity cost is

Note:

unit . Equations , with the stated data, give

yuan.

Thus, ideally, this heat pump costs about yuan for .

. Let , be the inner and outer air-layer surface temperatures. The rates of heat conduction through the inner glass, air and outer glass are respectively

In steady heat conduction,

Equations give

and

Solving gives

Substitute into :

Constant temperature requires . By , for double glazing the minimum electric power is

Ideally, the cost for operation time is

Substituting the data gives

yuan.